1 **(a)** Recognises that tangents from an external point are equal in length, so triangle PAB is isosceles: $180 - 44 = 136$ **[M1]**

$136 \div 2 = 68$ **[A1]**

(b) Recognises angle between tangent and radius is $90°$, setting up sum of angles in quadrilateral $OAPB$: $360 - 90 - 90 - 44$ **[M1]**

136 **[A1]**

[Total Marks: 4]

2 **(a)** Tangents from an external point are equal in length **[B1]**

(b) 70 **[B1]**

(c) Angle $OAP = 90°$ OR Angle $OBP = 90°$ **[M1]**

140 **[A1]**

[Total Marks: 4]

3 **(a)** Angle $PTO = 56 \div 2 = 28°$ **[M1]**

$180 - 90 - 28 = 62$ **[A1]**

(b) Tangents from an external point are equal in length (so $TP = TQ$) **[B1]**

(c) 62 **[B1]**

[Total Marks: 4]

4 **(a)** Uses angle sum of quadrilateral $OATB$ with angle $OAT = 90°$ and angle $OBT = 90°$: $360° - 90° - 90° - 44°$ OR $180° - 44°$ **[M1]**

136 **[A1]**

(b) Recognises triangle OAB is isosceles ($OA = OB$) and uses $(180° - 136°) \div 2$ **[M1]**

22 **[A1]**

(c) Tangents from an external point are equal in length **[B1]**

[Total Marks: 5]

5 **(a)** 90 **[B1]**

The perpendicular from the centre of a circle to a chord bisects the chord (or perpendicular bisector of a chord passes through the centre) **[B1]**

(b) Uses midpoint M: $AM = \dfrac{16}{2} = 8$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 8^2 = 10^2$ or $OM = \sqrt{10^2 - 8^2}$ **[M1]**

6 **[A1]**

[Total Marks: 5]

6 **(a)** 24 **[B1]**

(b) Identifies $\angle OAP = 90°$ (tangent perpendicular to radius) **[M1]**

$OP^2 = 7^2 + 24^2 = 625 \implies OP = 25$ **[A1]**

(c) Let M be the intersection of OP and AB. Identifies $\triangle OAM \sim \triangle OPA$ OR uses Area of $\triangle OAP$ OR uses $\cos(\angle AOP)$ **[M1]**

$\cos(\angle AOP) = \dfrac{7}{25}$ and $OM = 7 \times \cos(\angle AOP)$ **[M1]**

$OM = \dfrac{49}{25}$ or 1.96 **[A1]**

[Total Marks: 6]

7 **(a)** $AM = 12$ cm **[B1]**

Reason: the perpendicular from the centre of a circle to a chord bisects the chord **[B1]**

(b) $OM^2 = 13^2 - 12^2$ $(= 169 - 144 = 25)$ **[M1]**

5 **[A1]**

(c) $ON = 5$ cm **[B1]**

Reason: equal chords are equidistant from the centre of the circle **[B1]**

[Total Marks: 6]

8 **(a)** $x + 3 = 3x - 5$ **[M1]**

$x = 4$ **[A1]**

(b) Perpendicular distance = 7 (can be implied) **[B1]**

$\sqrt{25^2 - 7^2}$ OR 24 **[M1]**

48 **[A1]**

[Total Marks: 5]

9 **(a)** 12 **[B1]**

Tangents from an external point to a circle are equal in length **[B1]**

(b) Uses $\angle OAP = 90°$ and Pythagoras' theorem: $OP = \sqrt{5^2 + 12^2}$ **[M1]**

13 **[A1]**

(c) Calculates area of two congruent right-angled triangles: $2 \times \left(\dfrac{1}{2} \times 5 \times 12\right)$ **[M1]**

60 **[A1]**

[Total Marks: 6]

10 **(a)** $OA = OB$ (radii of the circle) **[B1]**

$PA = PB$ (tangents from an external point are equal in length) **[B1]**

OP is common to both triangles (or angle OAP = angle $OBP = 90°$, tangent to radius $= 90°$), so triangles are congruent (SSS or RHS) **[B1]**

(b) (i) Angle OAP = angle $OBP = 90°$ **[M1]**

Angle $AOB = 360° - 90° - 90° - 52° = 128$ **[A1]**

(ii) $OA = OB$ so triangle OAB is isosceles **[M1]**

Angle $OAB = \dfrac{180° - 128°}{2} = 26$ **[A1]**

(c) 14.3 **[B1]**

[Total Marks: 8]

11 (a) Angle $OAP = 90°$ AND Angle $OBP = 90°$ (tangent perpendicular to radius) **[B1]**

Angle $AOB = 360 - 90 - 90 - 46 = 134°$ **[B1]**

(b) Triangle OAB is isosceles ($OA = OB$) **[M1]**

Angle $OAB = (180 - 134) \div 2 = 23°$ **[A1]**

[Total Marks: 4]

12 (a) Angle $OMB = 90°$ (line from centre to midpoint of chord is perpendicular to chord) **[B1]**

$MB = 7$ **[B1]**

$OM = \sqrt{8.5^2 - 7^2}$ **[M1]**

4.77 or 4.769... **[A1]**

(b) Half chord $= \sqrt{8.5^2 - 3^2}$ **[M1]**

$2 \times \sqrt{63.25} = 15.9$ or 15.90... **[A1]**

[Total Marks: 5]

13 (a) Identifies right-angled triangle OAM (or OBM) and states $AM = 24 \div 2 = 12$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = \sqrt{25} = 5$ **[A1]**

(b) 24 (using property that chords equidistant from the centre are equal in length) **[B1]**

(c) $\dfrac{1}{2} \times 24 \times 5$ **[M1]**

60 **[A1]**

[Total Marks: 6]

14 (a) The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord. **[B1]**

(b) $AM = 24 \div 2 = 12$ cm **[B1]**

$OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = 5$ **[A1]**

(c) $\sin(\angle AOM) = \frac{12}{13}$ OR $\cos(\angle AOM) = \frac{5}{13}$ OR $\tan(\angle AOM) = \frac{12}{5}$ **[M1]**

$\angle AOM = 67.38...^\circ$ **[A1]**

Angle $AOB = 2 \times 67.38... = 134.8^\circ$ **[A1]**

[Total Marks: 7]

15 (a) States or uses $OM \perp AB$ (perpendicular bisector of chord passes through centre), so $AM = 7$ **[B1]**

Uses Pythagoras' Theorem: $OM = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ **[B1]**

(b) States $ON = OM = 24$ (equal chords are equidistant from centre) **[B1]**

Uses Cosine rule: $MN^2 = 24^2 + 24^2 - 2(24)(24)\cos 100^\circ$ OR uses $MN = 2 \times 24 \times \sin 50^\circ$ **[M1]**

36.8 (accept $36.77\ldots$) **[A1]**

(c) $\frac{1}{2} \times 24 \times 24 \times \sin 100^\circ$ **[M1]**

284 (accept $283.6\ldots$) **[A1]**

[Total Marks: 7]

16 (a) Forms a right-angled triangle with hypotenuse 13 and base $\frac{24}{2} = 12$ **[M1]**

Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

5 **[A1]**

(b) 5 **[B1]**

(c) Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

Half of chord $EF = 5$ **[A1]**

10 **[A1]**

(d) The perpendicular from the centre to a chord bisects the chord (accept "perpendicular bisector of a chord passes through the centre") **[B1]**

[Total Marks: 8]

17 (a) Equal chords are equidistant from the centre, so $OM = ON$. **[B1]**

$5x - 7 = 2x + 2$ **[M1]**

$3x = 9 \implies x = 3$ **[A1]**

(b) $OM = 5(3) - 7 = 8$ cm **[B1]**

Uses Pythagoras' theorem: $AM^2 + 8^2 = 17^2$ **[M1]**

$AM = \sqrt{289 - 64} = \sqrt{225} = 15$ **[A1]**

$AB = 2 \times 15 = 30$ **[A1]**

[Total Marks: 7]

18 **(a)** Angle $OTA = 90°$ (tangent perpendicular to radius) **[B1]**

$AT = 12$ (perpendicular from centre to chord bisects chord) **[B1]**

In triangle OTA, $r^2 + 12^2 = R^2$ (Pythagoras theorem) **[M1]**

$R^2 - r^2 = 144$ **[A1]**

(b) Area $= \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ **[M1]**

$k = 144$ **[A1]**

(c) Distance $= r$ (or same distance as chord AB) **[B1]**

Equal chords are equidistant from the centre **[B1]**

[Total Marks: 7]

19 **(a)** The perpendicular bisector of a chord passes through the centre O, so $OP \perp AB$ and $OQ \perp CD$ **[B1]**

Since AB and CD are parallel, both OP and OQ are perpendicular to the same parallel lines through O, forming a single straight line POQ **[B1]**

(b) Identifies $AP = 12$, $CQ = 5$, and $OQ = 17 - x$ **[B1]**

Sets up Pythagoras' equations for both triangles: $r^2 = x^2 + 12^2$ and $r^2 = (17 - x)^2 + 5^2$ **[M1]**

Equates expressions for r^2: $x^2 + 144 = 289 - 34x + x^2 + 25$ **[M1]**

Solves linear equation: $34x = 170$ giving $x = 5$ **[A1]**

Substitutes $x = 5$ to find $r = \sqrt{5^2 + 12^2} = 13$ **[A1]**

[Total Marks: 7]

20 **(a)** The perpendicular bisector of a chord passes through the centre of the circle **[B1]**

(b) $MX = 16 \div 2 = 8$ **[B1]**

$OX = \sqrt{10^2 - 8^2} = 6$ **[M1]**

$10 - 6 = 4$ **[A1]**

(c) Distance of PQ from $O = 11 - 6 = 5$ **[B1]**

Half length of $PQ = \sqrt{10^2 - 5^2}$ **[M1]**

$\sqrt{75}$ or 8.66... **[A1]**

$2 \times \sqrt{75} = 17.3$ (17.32...) **[A1]**

[Total Marks: 8]

21 **(a)** Identifies $\angle OAT = 90°$ and $\angle AOT = 90° - x$ **[M1]**

Uses $AM = r\sin(90° - x) = r\cos x$ and perpendicular bisector property $AB = 2AM$ to get $2r\cos x$ **[A1]**

(b) $r\sin x$ (equal chords are equidistant from centre, distance $= OM = r\cos(90° - x)$) **[B1]**

(c) Area of $OAB = \frac{1}{2} \times AB \times OM = \frac{1}{2}(2r\cos x)(r\sin x) = r^2 \sin x \cos x$ **[M1]**

$TM = \frac{AM}{\tan x} = \frac{r\cos x}{\tan x} = \frac{r\cos^2 x}{\sin x}$ **[M1]**

Area of $ATB = \frac{1}{2} \times AB \times TM = \frac{1}{2}(2r\cos x)\left(\frac{r\cos^2 x}{\sin x}\right) = \frac{r^2\cos^3 x}{\sin x}$ **[M1]**

Equates areas: $r^2 \sin x \cos x = \frac{r^2\cos^3 x}{\sin x} \implies \sin^2 x \cos x = \cos^3 x \implies \tan^2 x = 1$ **[M1]**

45 **[A1]**

[Total Marks: 8]

22 (a) Tangents from an external point are equal in length **[B1]**

(b) Recognises $BP = BQ$, $CR = CQ$, and $DR = DS$ **[M1]**

$AB + CD = AP + PB + CR + RD$ **[M1]**

Substitutes equal tangent segments:
$AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$ **[A1]**

(c) Substitutes expressions into the relationship from part (b):
$(2x + 5) + (x + 8) = (2x + 1) + (3x - 2)$ **[M1]**

Correctly expands and simplifies to a linear equation, e.g., $3x + 13 = 5x - 1$ **[M1]**

7 **[A1]**

(d) Substitutes $x = 7$ into the lengths of the sides (at least two) OR uses $2(3x + 13)$ **[M1]**

68 **[A1]**

[Total Marks: 9]

23 (a) Recognises angle $OPT = 90°$ and sets up Pythagoras: $TP^2 + 7^2 = 25^2$ **[M1]**

$TP = \sqrt{625 - 49} = \sqrt{576} = 24$ **[A1]**

(b) Uses trigonometry in triangle OPT: $\sin(\text{angle } PTO) = \frac{7}{25}$ or $\tan(\text{angle } PTO) = \frac{7}{24}$ **[M1]**

Angle $PTO = 16.26°$ **[A1]**

Angle $PTQ = 2 \times 16.26° = 32.5°$ (accept $32.52°$) **[A1]**

(c) Calculates total area of kite $OPTQ = 2 \times (\frac{1}{2} \times 7 \times 24) = 168$ **[M1]**

Calculates angle $POQ = 360 - 90 - 90 - 32.52 = 147.48°$ (or uses $180 - 32.52$) **[M1]**

Calculates area of sector $OPQ = \frac{147.48}{360} \times \pi \times 7^2 = 63.07\ldots$ **[M1]**

Shaded area $= 168 - 63.07 = 104.93$ (accept 104.9 to 105.0) **[A1]**

[Total Marks: 9]

24 **(a)** $AM = 8$ cm and $CN = 6$ cm (perpendicular from centre bisects chord), giving $r^2 = x^2 + 8^2$ **[B1]**

$r^2 = (14 - x)^2 + 6^2$ **[B1]**

$x^2 + 64 = 196 - 28x + x^2 + 36$ leading to $28x = 168$ and $x = 6$ **[M1]**

$r = \sqrt{6^2 + 8^2} = 10$ **[A1]**

(b) Angle $OAT = 90°$ (tangent to radius $= 90°$) and $\tan(\text{angle } AOM) = \dfrac{8}{6}$ **[M1]**

$TA = 10 \times \dfrac{8}{6}$ (or similar triangles OMA and OAT used correctly) **[M1]**

$\dfrac{40}{3}$ **[A1]**

(c) $TB = \dfrac{40}{3}$ cm (or 13.3 cm) **[B1]**

Perimeter $= 10 + 10 + \dfrac{40}{3} + \dfrac{40}{3} = 46.7$ cm **[B1]**

(d) $\tan(\text{angle } ATO) = \dfrac{10}{40/3} = \dfrac{3}{4}$ giving angle $ATO = 36.87°$ **[M1]**

Angle $ATB = 2 \times 36.87° = 73.7$ **[A1]**

[Total Marks: 11]